



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

TG: @Chalnaayaaar

MINOR

04-03-2023

ENTHUSIAST ADVANCE COURSE PHASE : MEC-I (B)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

This Booklet contains 28 pages.

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on the Back Cover of this Test Booklet.

Important Instructions :

1. On the Answer Sheet, fill in the particulars on **Side-1** and **Side-2** carefully with **blue/black** ball point pen only.
2. The test is of **3 Hours 20 Minutes** duration and this Test Booklet contains **200** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. In this Test Paper, each subject will consist of **two sections**. **Section A** will consist of **35** questions (all questions are mandatory) and **Section B** will have **15** questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.
4. In case of more than one option correct in any question, the best correct option will be considered as answer.
5. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. **On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.**
8. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
9. Use of white fluid for correction is **not** permissible on the Answer Sheet.

Name of the Candidate (in Capitals) _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in Pre-Medical 2024

Topic : GRAVITATION, NLM, FRICTION, ELECTROSTATICS (TILL DATE), UNIT, DIMENSIONS, ERROR & MEASUREMENT

SECTION - A

Attempt All 35 questions

1. Let the minimum external work done in shifting a particle from center of earth to earth's surface be w_1 and that from surface of earth to infinity be w_2 . Then $\frac{w_1}{w_2}$ is equal to :-

- (1) 1:1 (2) 1:2
(3) 2:1 (4) 1:3

2. A body is projected up from the surface of the earth with a velocity equal to $\frac{3}{4}$ th of escape velocity for earth's surface. If R be the radius of the earth, the maximum height it reaches is :-

- (1) $\frac{3R}{10}$ (2) $\frac{9R}{7}$ (3) $\frac{8R}{5}$ (4) $\frac{9R}{5}$

3. How much deep inside the earth (radius R) should a man go, so that his weight becomes one fourth of that on the earth's surface ?

- (1) $\frac{R}{4}$ (2) $\frac{R}{2}$
(3) $\frac{3R}{4}$ (4) None of these

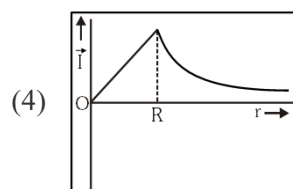
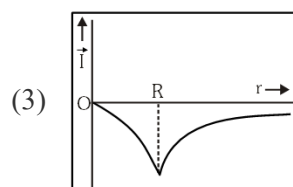
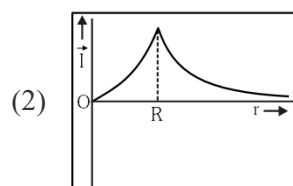
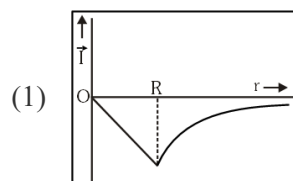
4. The satellite of mass ' m ' revolving in a circular orbit of radius ' r ' around the earth has kinetic energy E . then its angular momentum about the center of earth is :-

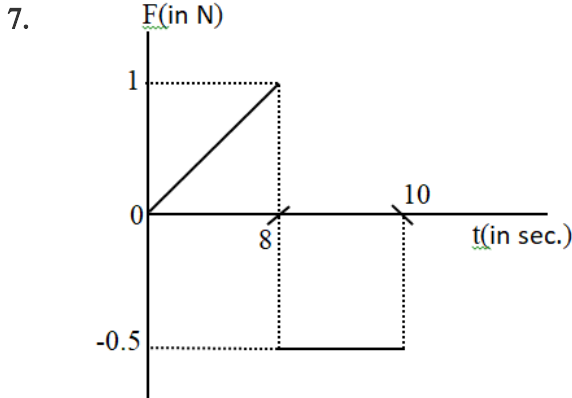
- (1) $\sqrt{\frac{E}{mr^2}}$ (2) $\frac{E}{2mr^2}$
(3) $\sqrt{2Emr^2}$ (4) $\sqrt{2Emr}$

5. A particle of mass m is moving in a horizontal circle of radius R under a centripetal force equal to $-\frac{A}{r^2}$ ($A = \text{constant}$). The total energy of the particle is :-
(Potential energy at very large distance is zero)

- (1) $\frac{A}{R}$ (2) $-\frac{A}{R}$
(3) $\frac{A}{2R}$ (4) $-\frac{A}{2R}$

6. Following curve shows the variation of intensity of gravitational field ($\vec{I}$) with distance from the centre of solid sphere(r) :



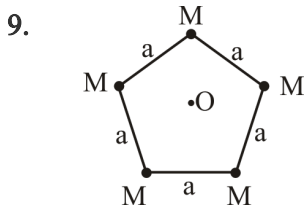


A force - time graph for the motion of a body is shown in the figure. The change in the momentum of the body between 0 to 10 sec. is -

- (1) Zero (2) $4 \frac{\text{kg m}}{\text{sec.}}$
 (3) $5 \frac{\text{kg m}}{\text{sec.}}$ (4) $3 \frac{\text{kg m}}{\text{sec.}}$

8. At some planet 'g' is 1.96 m/sec^2 . If it is safe to jump from a height of 2m on earth, then what should be corresponding safe height for jumping on that planet:-

- (1) 5 m
 (2) 2 m
 (3) 10 m
 (4) 20 m



Find net gravitational field intensity at O.

- (1) $\frac{GM}{a^2}$ (2) $\frac{2GM}{a^2}$
 (3) 0 (4) $\frac{2\sqrt{3}GM}{a^2}$

10. A body of mass 'm' is lifted up from earth's surface to a height 3 times the radius of earth. Change in potential energy of the body is : ('g' is acceleration due to gravity at earth's surface)

- (1) $3mgR$ (2) $\frac{3}{4}mgR$
 (3) $\frac{1}{3}mgR$ (4) $\frac{2}{3}mgR$

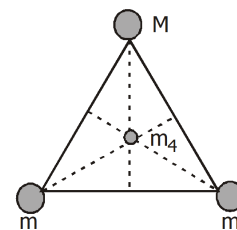
11. Two satellites A and B of masses m_1 and m_2 ($m_1 = 2m_2$) are moving in circular orbits of radii r_1 and r_2 ($r_1 = 4r_2$) respectively, around the earth. If their time period are T_A and T_B , then the ratio T_A/T_B is :-

- (1) 4 (2) 8 (3) 2 (4) 16

12. The radius of the earth is R. The height of a point vertically above the earth's surface at which acceleration due to gravity becomes 1% of its value at the surface is:-

- (1) 8 R (2) 9 R
 (3) 10 R (4) 20 R

13. Two spheres of mass m and a third sphere of mass M form an equilateral triangle, and a fourth sphere of mass m_4 is at the center of the triangle. If the net gravitational force on m_4 from the three other spheres is zero, then :



- (1) $M = m$ (2) $m = m/\sqrt{3}$
 (3) $M = \sqrt{3}m$ (4) $M = \sqrt{2}m$

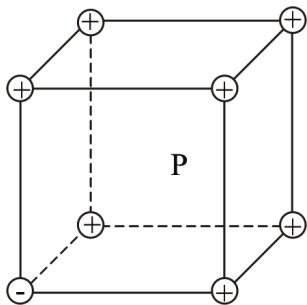
14. Assume that a tunnel is dug through earth from North pole to south pole and that the earth is a non-rotating, uniform sphere of density ρ . The gravitational force on a particle of mass m dropped into the tunnel when it reaches a distance r from the centre of earth is

(1) $\left(\frac{3}{4\pi}mG\rho\right)r$ (2) $\left(\frac{4\pi}{3}mG\rho\right)r$
 (3) $\left(\frac{4\pi}{3}mG\rho\right)r^2$ (4) $\left(\frac{4\pi}{3}m^2G\rho\right)r$

15. Electric field due to a point charge Q at a distance r is E_1 . If same charge is uniformly distributed on semicircular arc of radius r . Then electric field at centre is E_2 . Then :-

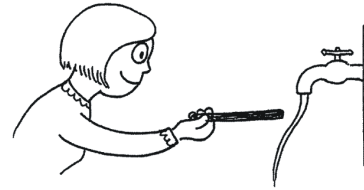
(1) $E_1 < E_2$ (2) $E_1 > E_2$
 (3) $E_1 = E_2$ (4) $E_2 = 0$

16. Consider a regular cube with positive point charge $+Q$ in all corners except for one which has a negative point charge $-Q$. Let the distance from any corner to the center of the cube be r . What is the magnitude of electric field at point P , the center of the cube?



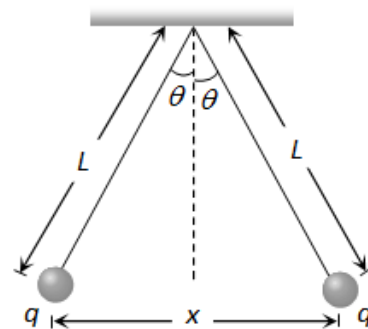
(1) $E = 0$
 (2) $E = 1 \text{ kQ}/r^2$
 (3) $E = 2 \text{ kQ}/r^2$
 (4) $E = 6 \text{ kQ}/r^2$

17. In normal cases thin stream of water bends toward a negatively charged rod. When a positively charged rod is placed near the stream, it will bend in the :-



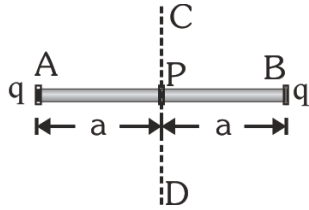
- (1) Opposite direction
 (2) Same direction.
 (3) It won't bend at all
 (4) Can't be predicted.

18. In the given figure two tiny conducting balls of identical mass m and identical charge q hang from non-conducting threads of equal length L . Assume that θ is so small that $\tan\theta \approx \sin\theta$, then for equilibrium x is equal to



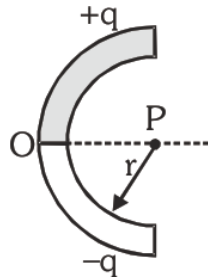
(1) $\left(\frac{q^2L}{2\pi\epsilon_0mg}\right)^{\frac{1}{3}}$ (2) $\left(\frac{qL^2}{2\pi\epsilon_0mg}\right)^{\frac{1}{3}}$
 (3) $\left(\frac{q^2L^2}{4\pi\epsilon_0mg}\right)^{\frac{1}{3}}$ (4) $\left(\frac{q^2L}{4\pi\epsilon_0mg}\right)^{\frac{1}{3}}$

19. For the arrangement shown in the figure below, mark out the correct statement.



- (1) For +ve test charge, P is a position of stable equilibrium for displacement along AB.
- (2) For +ve test charge, P is a position of unstable equilibrium for displacement along CD.
- (3) For -ve test charge, P is a position of unstable equilibrium for displacement along AB.
- (4) All of the above

20. A thin glass rod is bent into a semicircle of radius r . A charge $+q$ is uniformly distributed along the upper half and a charge $-q$ is uniformly distributed along the lower half, as shown in the figure. The magnitude and direction of the electric field $\vec{E}$ produced at P, the centre of the circle, will be :-



- (1) 0
- (2) $\frac{q}{\epsilon_0 \pi^2 r^2}$ perpendicular to the line OP and directed downward
- (3) $\frac{q}{\epsilon_0 \pi r^2}$ perpendicular to the line OP and directed downward
- (4) $\frac{q}{\epsilon_0 \pi r^2}$ along the axis OP

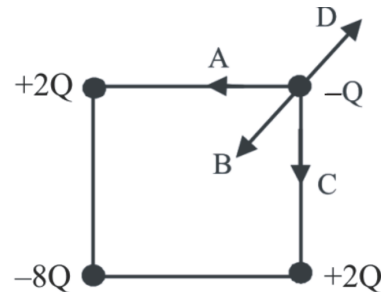
21. A disc of radius R is charged on its surface with surface charged density $\sigma = \sigma_0 r^3$. Here σ_0 is a constant and r is distance from its centre. Total charge on the disc is :-

- (1) $\frac{2\pi\sigma_0 R^5}{5}$
- (2) $2\pi\sigma_0 R^5$
- (3) $\frac{\pi\sigma_0 R^4}{2}$
- (4) $\frac{\sigma_0 R^4}{4}$

22. In a medium, the force of attraction between two point charges, distance d apart, is F . What distance apart should these point charges be kept in the same medium, so that the force between them becomes $16F$?

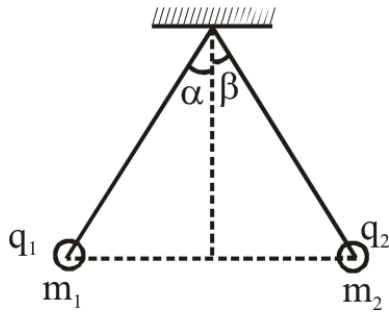
- (1) $\frac{d}{2}$
- (2) $\frac{d}{4}$
- (3) $\frac{d}{\sqrt{3}}$
- (4) $\frac{d}{\sqrt{2}}$

23. Four point charges with varying signs are arranged at the corners of a square as shown. Which of the arrows labeled A, B, C, and D gives the correct direction of the net force that acts on the charge at the upper right corner?

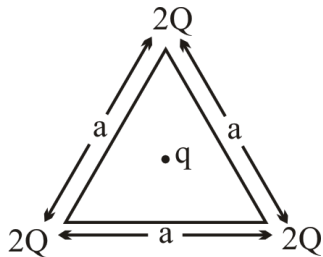


- (1) B
- (2) C
- (3) D
- (4) The net force on that charge is zero

24. Consider the following suspended charged ball system. If $\alpha > \beta$ then at equilibrium which of the following may be true :-



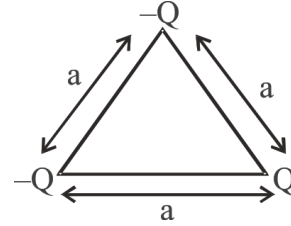
- (1) $q_1 > q_2$ & $m_1 = m_2$
 (2) $q_1 = q_2$ & $m_1 = m_2$
 (3) $q_1 > q_2$ & $m_1 < m_2$
 (4) $q_1 = q_2$ & $m_1 > m_2$
25. Find the value of q such that whole system remains in equilibrium ?



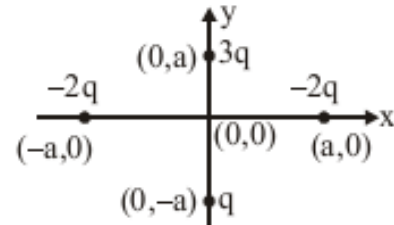
- (1) Q (2) $-\frac{2Q}{\sqrt{3}}$
 (3) $\frac{Q}{\sqrt{3}}$ (4) $-\frac{Q}{\sqrt{3}}$
26. Find max. electric field intensity at axial point of uniformly charge ring of radius "R" & charge "Q":

- (1) $\frac{2}{3\sqrt{3}} \frac{KQ}{R^2}$ (2) $\frac{1}{3\sqrt{3}} \frac{KQ}{R^2}$
 (3) $\frac{3}{2\sqrt{3}} \frac{KQ}{R^2}$ (4) $3\sqrt{2} \frac{KQ}{R^2}$

27. Find net force on charge Q : TG: @Chalnaayaaar

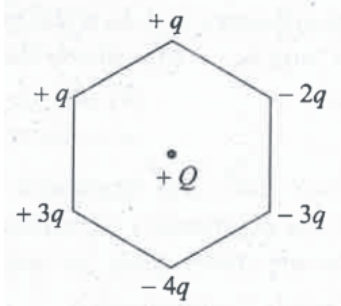


- (1) $\frac{KQ^2}{a^2}$
 (2) $-\frac{KQ^2}{a^2}$
 (3) $\frac{\sqrt{3} KQ^2}{a^2}$
 (4) $-\frac{\sqrt{3} KQ^2}{2a^2}$
28. An electron is moving in circular path of radius r , around a proton. The time period of proton is directly propotional to :-
- (1) r^2 (2) r^3 (3) $r^{3/2}$ (4) r^0
29. 4 charges are placed each at a distance 'a' from the origin. Electric field intensity at origin is :-



- (1) $\frac{-2Kq}{a^2} \hat{j}$
 (2) $\frac{2Kq}{a^2} \hat{j}$
 (3) $\frac{Kq}{a^2} \hat{j}$
 (4) None

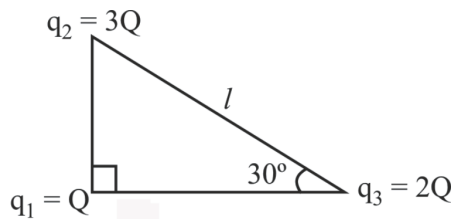
30. Six charges are kept at the vertices of a regular hexagon as shown in the figure.



The magnitude of force applied by +Q charge on +q charge is F. If the net electric force on +Q charge is nF then find the value of n.

- (1) 4 (2) 5 (3) 7 (4) 9

31. For the given figure, if F_1 is force on q_1 due to q_2 and F_2 is force on q_1 due to q_3 , then find $\frac{F_1}{F_2}$.



- (1) $\frac{7}{2}$ (2) $\frac{3}{2}$ (3) $\frac{3\sqrt{3}}{2}$ (4) $\frac{9}{2}$

32. The no. of significant figures in 0.003208 is

- (1) 5 (2) 4 (3) 3 (4) 2

33. The dimensions of electric potential is :

- (1) $M^1 L^1 T^{-2} A^{-2}$ (2) $M^1 L^2 T^{-3} A^{+2}$
(3) $M L^2 T^{-3} A^{-1}$ (4) $M^1 L^2 T^{-4} A^{-1}$

34. A wave is represented by

$$y = a \sin (At - Bx + C)$$

where A, B, C are constants. The dimensions of A, B, C are

- (1) $T^{-1}, L, M^0 L^0 T^0$ (2) $T^{-1}, L^{-1}, M^0 L^0 T^0$
(3) T, L, M (4) T^{-1}, L^{-1}, M^{-1}

35. The potential energy of a particle varies with distance x from a fixed origin as $U = \frac{A\sqrt{x}}{x^2 + B}$, where A and B are dimensional constants then dimensional formula for AB is :-

- (1) $[ML^{7/2} T^{-2}]$ (2) $[ML^{11/2} T^{-2}]$
(3) $[M^2 L^{9/2} T^{-2}]$ (4) $[M^{13/2} T^{-3}]$

SECTION - B

This section will have 15 questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

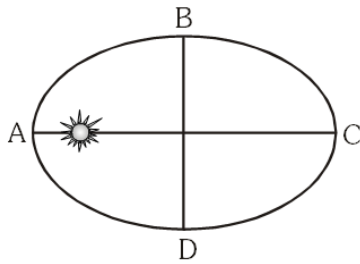
36. The time period of moon's revolution around the earth is approximately 29 days. If moon's mass were 2 fold its present value and all other things remain unchanged, the time period of moon's revolution would be nearly :-

- (1) $29\sqrt{2}$ days (2) $\frac{29}{\sqrt{2}}$ days
(3) 29×2 days (4) 29 days

37. Acceleration due to gravity on moon is $\frac{1}{6}$ th of the acceleration due to gravity on earth. If the ratio of densities of earth (ρ_e) and moon (ρ_m) is $\frac{5}{3}$ (i.e. $\frac{\rho_e}{\rho_m} = \frac{5}{3}$) then radius of moon (R_m) in terms of R_e will be :-

- (1) $\frac{5}{18} R_e$ (2) $\frac{1}{6} R_e$
(3) $\frac{7}{18} R_e$ (4) $-\frac{1}{2\sqrt{3}} R_e$

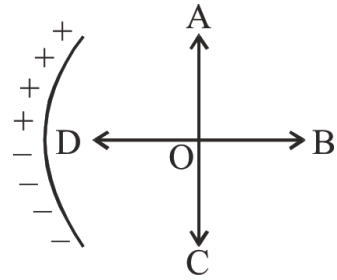
38. In adjoining figure earth goes around the sun in elliptical orbit on which point the orbital speed is maximum :



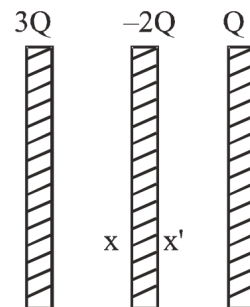
- (1) On A (2) On B
(3) On C (4) On D
39. A body of mass 5 kg starts from the origin with an initial velocity $\vec{u} = (30\hat{i} + 40\hat{j})$ m/s. if a constant force $\vec{F} = -(\hat{i} + 5\hat{j})$ N acts on the body, the time in which the y component of the velocity becomes zero is :-
- (1) 5 sec. (2) 20 sec.
(3) 40 sec. (4) 80 sec.
40. A tunnel is dug along the diameter of the earth (radius R and mass M). There is a particle of mass 'm' at the centre of the tunnel. The minimum velocity given to the particle so that it just reaches to the surface of the earth is :-

- (1) $\sqrt{\frac{GM}{R}}$
(2) $\sqrt{\frac{GM}{2R}}$
(3) $\sqrt{\frac{2GM}{R}}$
(4) It will reach with the help of a negligible velocity

41. The linear charge density on upper half of semi-circular section of ring is λ and that at lower half is $-\lambda$. The direction of electric field at centre O of ring is :

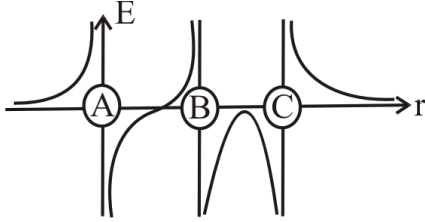


- (1) along OA (2) along OB
(3) along OC (4) along OD
42. Two identical balls equally charged, density 16 kg/m^3 suspended by string from common point are in equilibrium the angle between strings is 2θ . If whole system dip in liquid of density of 8 kg/m^3 at equilibrium angle remain same dielectric constant of liquid.
- (1) 3 (2) 2 (3) 4 (4) 8
43. Charge on both surfaces of middle metal plate :

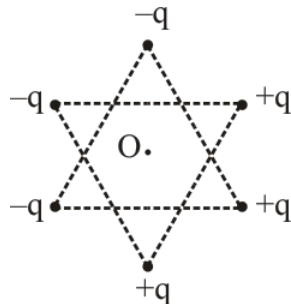


- (1) $x = -2Q, x' = 0$
(2) $x = +2Q, x' = 0$
(3) $x = -2Q, x' = +4Q$
(4) $x = -4Q, x' = +2Q$

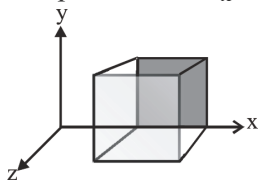
44. Identify the charges A, B and C



- (1) A positive B negative C negative
 (2) A negative B positive C negative
 (3) A negative B negative C positive
 (4) A positive B positive C negative
45. Electric force on charge q_0 at centroid O due to given charge distribution formed by two triangle is (where side of each triangle is a) :-



- (1) $\frac{6kqq_0}{a^2}$ (2) $\frac{12kqq_0}{a^2}$
 (3) $4\sqrt{3}\frac{kqq_0}{a^2}$ (4) zero
46. A cube of side a is placed such that the nearest face which is parallel to the $y-z$ plane is at a distance ' a ' from the origin. The electric field components are $E_x = \alpha x^{1/2}$, $E_y = E_z = 0$.



The flux ϕ_E through the cube is :-

- (1) $\sqrt{2}\alpha a^{5/2}$ (2) $-\alpha a^{5/2}$
 (3) $(\sqrt{2} - 1)\alpha a^{5/2}$ (4) zero

47. A solid metallic sphere has a charge $+3Q$. Concentric with this sphere is a conducting spherical shell having charge $-Q$. The radius of the sphere is ' a ' and that of the spherical shell is ' b ' ($b > a$). What is the electric field at a distance R ($a < R < b$) from the centre ?

- (1) $\frac{4Q}{2\pi\epsilon_0 R^2}$
 (2) $\frac{3Q}{4\pi\epsilon_0 R^2}$
 (3) $\frac{3Q}{2\pi\epsilon_0 R^2}$
 (4) $\frac{Q}{2\pi\epsilon_0 R}$

48. Two charges $8q$ and q are placed at a distance ℓ apart. Another charged particle Q is placed in between them at mid-point. If the resultant force on q is zero, then the value of Q is :-

- (1) q (2) $-q$ (3) $2q$ (4) $-2q$

49. In a vernier callipers, one main scale division is x cm and n divisions of vernier scale coincide with $(n - 1)$ divisions of main scale. The least count (in cm) of the calliper is :

- (1) $\frac{(n - 1)}{n}x$ (2) $\frac{nx}{(n - 1)}$
 (3) $\frac{x}{n}$ (4) $\frac{x}{(n - 1)}$

50. A student measure quantities a, b and c and then calculates S by the formula $S = ab^2/c^3$. If the errors in a, b, c are 1%, 3% and 2% respectively, the maximum error in S can be:-

- (1) 13% (2) 7%
 (3) 4% (4) 1%

Topic : CHEMICAL KINETICS, SOLUTION (TILL CONCENTRATION TERMS), NOMENCLATURE, ISOMERISM

SECTION-A

Attempt All 35 questions

51. For a first order reaction $A(g) \rightarrow B(g)$, after 64 minutes $\frac{1}{16}$ th of initial amount of reactant is left, then rate constant is

- (1) $\frac{\ln 2}{4} \text{ min}^{-1}$ (2) $\frac{\ln 2}{8} \text{ min}^{-1}$
(3) $\frac{\ln 2}{16} \text{ min}^{-1}$ (4) $\frac{\ln 2}{64} \text{ min}^{-1}$

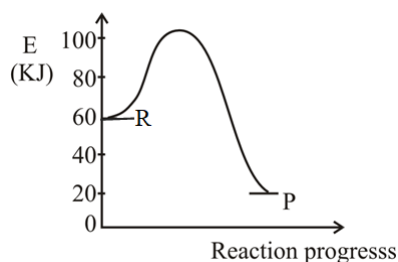
52. For a reaction $A \rightarrow B$, it was found that $t_{1/2} \times [A]_0^2 = \text{constant}$ then, it is a :

- (1) zero order reaction
(2) first order reaction
(3) second order reaction
(4) complex reaction

53. Rate of a reaction at 10°C is r , then upon increasing temperature to 50°C , rate will be –

- (1) $32r$ (2) $16r$ (3) $4r$ (4) $8r$

54. For a reaction $R \rightarrow P$,



then select the correct answer –

- (1) $E_a = 60 \text{ KJ/mol}$ (2) $\Delta H = -40 \text{ KJ/mol}$
(3) $E_a = 40 \text{ KJ/mol}$ (4) Both (2) and (3)

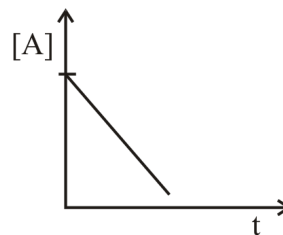
55. For a particular reaction, dependence of rate constant on temperature may be given as $\log k = 5 - \frac{200K}{T}$, where K is in sec^{-1} , then value of A is –

- (1) 10^5 sec^{-1}
(2) 200 sec^{-1}
(3) $2.303 \times 200 \text{ sec}^{-1}$
(4) $200 \times 2 \text{ sec}^{-1}$

56. A first order reaction is 40% completed in 20 minutes. How long will it take for 60% completion ?

- (1) 30 min. (2) 36.3 min.
(3) 40 min. (4) 45 min.

57. For reaction $A \rightarrow B$ if



then order of reaction is

- (1) 1 (2) 0 (3) 2 (4) 3

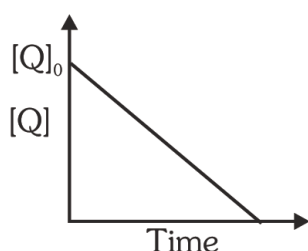
58. For reaction $2\text{N}_2\text{O}_5(g) \rightarrow 4\text{NO}_{2(g)} + \text{O}_{2(g)}$, $r = k[\text{N}_2\text{O}_5]^1$, then select the correct statement.

- (1) It is bimolecular reaction
(2) Half life is inversely proportional to $[\text{N}_2\text{O}_5]_0$
(3) rate of reaction is constant
(4) Rate constant is independent of $[\text{N}_2\text{O}_5]$

59. Half lives of first order and a zero order reaction are same. Then, the ratio of initial rates of first order reaction to that of the zero order reaction is (Given : $[A]_0 = 1\text{M}$)

- (1) $\frac{0.693}{2}$ (2) 4×0.693
(3) $2 \ln 2$ (4) $\ln 8$

60. In the reaction : $P + Q \rightarrow R + S$
the time taken for 75% conversion of P is twice the time taken for 50% conversion of P. The concentration of Q varies with reaction time as shown in the figure. The overall order of the reaction is



- (1) 2 (2) 3 (3) 0 (4) 1
61. For the reaction : $A_2 + 2B \rightarrow 2AB$, the following rate data is obtained

$[A_2]$	$[B]$	$-\frac{d[A_2]}{dt}$
0.1	0.2	$1 \times 10^{-2} \text{ M/s}$
0.2	0.2	$2 \times 10^{-2} \text{ M/s}$
0.2	0.4	$8 \times 10^{-2} \text{ M/s}$

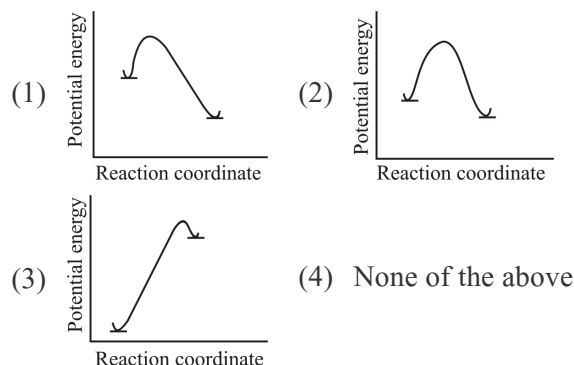
The order of reaction w.r.t. A_2 & B are respectively

- (1) 1, 2 (2) 2, 1 (3) 1, 0 (4) 0, 2
62. According to Arrhenius equation, a straight line is to be obtained by plotting the logarithm of the rate constant of chemical reaction ($\log k$) against :
- (1) T (2) $\log T$
(3) $\log (1/T)$ (4) $1/T$

63. An endothermic reaction $A \rightarrow B$ have an activation energy 15 kcal/mol and the enthalpy of the reaction is 5 kcal/mol. The activation energy of the reaction $B \rightarrow A$ is :-

- (1) 20 kcal/mol (2) 15 kcal/mol
(3) 10 kcal/mol (4) Zero

64. An endothermic reaction with high activation energy for the forward reaction is given by the diagram :



65. The rate constant for a reaction at the initial concentration of $2 \times 10^{-2} \text{ mol L}^{-1}$ is $4 \times 10^{-3} \text{ mol L}^{-1} \text{ min}^{-1}$. The half-life period for this reaction in seconds is :-

- (1) 300 (2) 150 (3) 180 (4) 240

66. In a gaseous phase reaction : $A_2(g) \rightarrow B(g) + (1/2)C(g)$, the increase in pressure from initial pressure of 100 mm to 120 mm is noticed in 5 minute. The rate of disappearance of A_2 in mm min^{-1} is :

- (1) 4 (2) 8 (3) 16 (4) 2

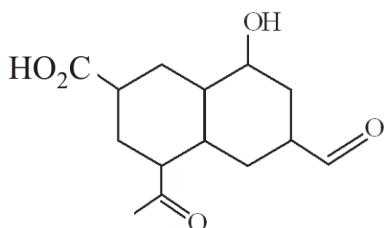
67. The molarity of 648 g of pure water is :-

- (1) 36 M (2) 55.5 M
(3) 3.6 M (4) 5.55 M

68. What is the concentration of sugar ($C_{12}H_{22}O_{11}$) in mol/L if 34.2 g sugar is dissolved in enough water to make final volume upto 2L?

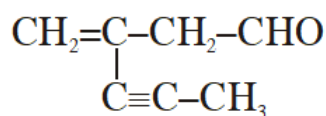
- (1) 0.01 M (2) 0.05 M
(3) 0.02 M (4) 0.5 M

69. How many different functional groups are present in given compound ?



- (1) 3 (2) 5 (3) 4 (4) 2

70. How many carbon atoms are present in principle carbon chain of following compound?



- (1) 7 (2) 6
(3) 5 (4) None

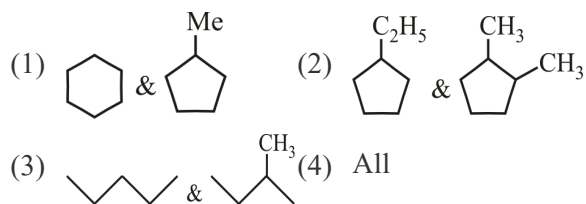
71. How many structural isomers of alkene are possible from C_5H_{10} ?

- (1) 10 (2) 5
(3) 4 (4) 8

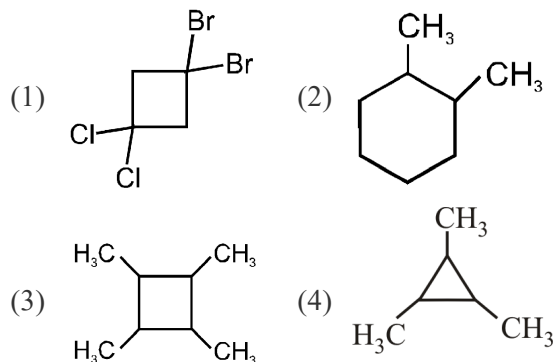
72. $(CH_3)_3C-CH=CH_2$ has the IUPAC name

- (1) 3,3-Dimethyl-but-1-ene
(2) 2,2-Dimethyl-but-1-ene
(3) 2,2-Dimethyl-but-3-ene
(4) 1,3-Dimethyl-propene

73. A pair of chain isomers is: TG: @Chalnaayaaar



74. Which of the following compound can not show geometrical isomerism?

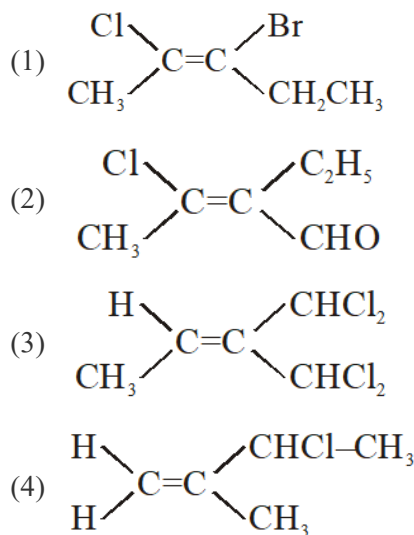


75. Calculate total number of geometrical isomer of given compound.

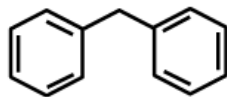


- (1) 4 (2) 3 (3) 2 (4) 1

76. Select the E-isomer among the following

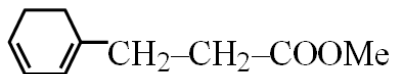


77. Number of 1° C, 2° C and 3° C in given compound is, respectively.

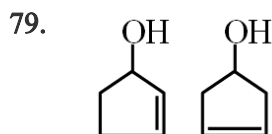


(1) 0,2,1 (2) 0,11,2 (3) 0,10,2 (4) 1,2,0

78. How many sp^2 - carbon atoms are present in given compound



(1) 7 (2) 5 (3) 4 (4) 3



Statement 1 $\rightarrow$ Both compounds are positional isomers of each other.

Statement 2 $\rightarrow$ Both compounds have different position of double bonds.

- (1) Statement 1 is correct and statement 2 is wrong.
 (2) Statement 1 is wrong and Statement 2 is correct.
 (3) Both Statements are correct
 (4) Both Statements are wrong.

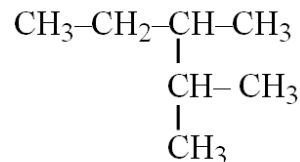
80. Select incorrect match

	Radical	Name
(1)	$\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2-$	Isobutyl
(2)	$\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-$	Secondary propyl
(3)	$\text{CH}_3-\text{CH}_2-\underset{\text{CH}_3}{\text{CH}}-$	Secondary butyl
(4)	$\text{CH}_2 = \text{CH}-$	Vinyl

81. Which of the following can show chain isomerism ?

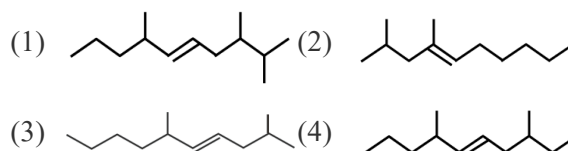
(1) C_3H_8 (2) C_3H_6
 (3) C_4H_{10} (4) C_2H_2

82. Correct IUPAC name of given compound is



- (1) 3 - Isopropyl butane
 (2) 2 - Isopropyl butane
 (3) 2,3 - Dimethyl pentane
 (4) 3,4 - Dimethyl pentane

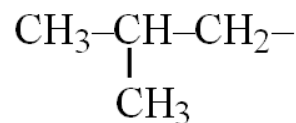
83. What is structure of 2, 6 - Dimethyl dec - 4 - ene



84. Which of the following show geometrical isomerism ?

(1) $\text{CH}_3-\text{CH}=\text{CH}_2$ (2) $\text{CH}_3-\text{CH}=\underset{\text{CH}_3}{\text{CMe}}$
 (3) $\text{CH}_3-\text{CH}=\text{CH}-\text{C}_2\text{H}_5$ (4) All of above

85. Correct IUPAC name of given radical is



- (1) Butyl (2) 2 - Methyl propyl
 (3) 2 - Methyl propane (4) 2 - Ethyl propyl

SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

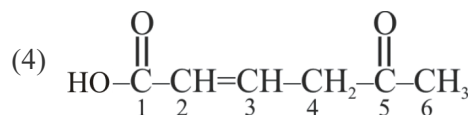
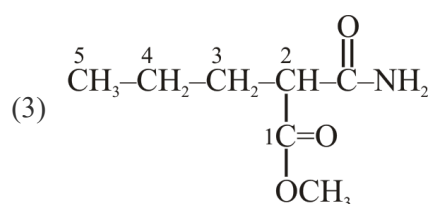
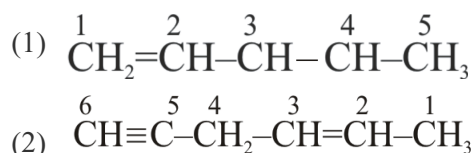
86. In a reaction $2A \rightarrow P$, concentration of A decreases from 0.5 mol L^{-1} to 0.4 mol L^{-1} in 10 minutes then rate of reaction is
 (1) $10^{-2} \text{ mol L}^{-1} \text{ min}^{-1}$ (2) $5 \times 10^{-3} \text{ mol L}^{-1} \text{ min}^{-1}$
 (3) $5 \times 10^{-3} \text{ mol L}^{-1} \text{ sec}^{-1}$ (4) $5 \times 10^{-2} \text{ mol L}^{-1} \text{ min}^{-1}$
87. Select the correct statement ?
 (1) Rate constant is temperature independent
 (2) For fast reaction, rate constant is very small.
 (3) For zero order reaction rate constant is equal to the rate of reaction.
 (4) For first order reaction, rate constant decreases with increase of concentration of reactant.
88. For reaction $A(g) \rightarrow 2B(g)$, if $K = 2 \times 10^{-2} \text{ atm sec}^{-1}$, and $P_{A(\text{initial})} = 0.5 \text{ atm}$, after 10 sec pressure of B is :
 (1) 0.4 atm (2) 0.2 atm
 (3) 0.1 atm (4) 0.3 atm
89. For an elementary reaction $A(g) + 2B(g) \rightarrow C(g)$, upon increasing the volume of container 3 times, rate of reaction -
 (1) increases 3 times
 (2) decreases 3 times
 (3) increases 27 times
 (4) decreases 27 times

90. At 100°C , the gaseous reaction $A \rightarrow 2B + C$ is found to be first order. Starting with pure A, if at the end of 10 minutes, the total pressure of the system is 160 mm and after very long time, it was 300 mm. The partial pressure of A at the end of 10 min is :-
 (1) 70 mm (2) 160 mm
 (3) 60 mm (4) 80 mm

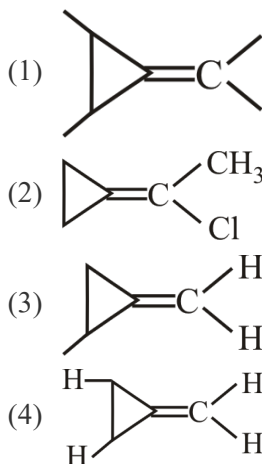
91. When initial concentration of a reactant is doubled in a reaction, its half-life period is not affected. The order of the reaction is :
 (1) First
 (2) Second
 (3) More than zero but less than first
 (4) Zero

92. 49 g of H_2SO_4 is present in 45g of H_2O find mole fraction of H_2O :-
 (1) $\frac{1}{6}$ (2) $\frac{1}{2}$ (3) $\frac{1}{3}$ (4) $\frac{5}{6}$

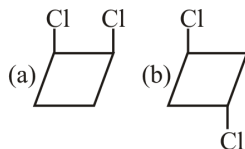
93. Which of the following system has incorrect IUPAC numbering :



94. Which of the following show geometrical isomerism



95. Consider following structures and select correct statement.



- (1) (a) and (b) are geometrical isomers of each other.
- (2) (a) has trans configuration and (b) has cis configuration
- (3) (a) and (b) are identical
- (4) (a) has cis configuration and (b) has trans configuration

96. Which category of compound does not show metamerism?

- (1) ether
- (2) ester
- (3) secondary amine
- (4) alcohol

97. Select correct statement(s) about conformational isomerism of ethane.

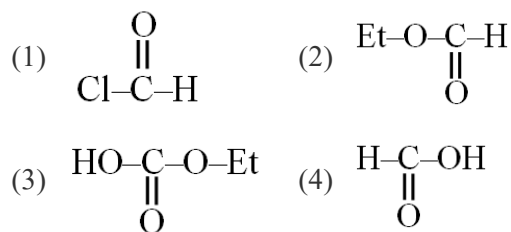
- (a) It has infinite number of conformation are possible.
- (b) Bond length and bond angle remain unchanged
- (c) Energy difference between maximum and minimum stable conformation is 12.5 kJ/mol.

(1) a, b (2) a, b, c (3) b, c (4) a only

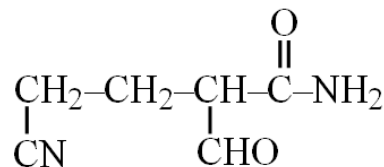
98. Correct IUPAC name of 2-Ethylbutan - 3 - ol

- (1) 3 - Ethylbutane - 2 - ol
- (2) 3 - Ethyl pentan - 2 - ol
- (3) 2 - Ethyl pentan - 3 - ol
- (4) 3-Methylpentan-2-ol

99. Which of the following has ester as P.F.G.



100. Correct IUPAC name of given compound is



- (1) 3-Cyano-2-formyl propanamide
- (2) 2 - Formyl - 4 - Cyano butanamide
- (3) 4 - Cyano - 2 - formyl butanamide
- (4) 4 - Cyano - 2 - formyl butane carboxamide

Topic : HUMAN HEALTH & DISEASE, ECOLOGY (UPTO INTERSPECIFIC INTERACTIONS)

SECTION-A

Attempt All 35 questions

101. **Assertion:** Malignant malaria is most serious one and can even be fatal.

Reason: Malignant malaria caused by Plasmodium vivax.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

102. Which one of the following pair of disease can spread through blood transfusion?

(1) AIDS and Typhoid Fever

(2) AIDS and Hepatitis-A

(3) Diabetes mellitus and syphilis

(4) Hepatitis B and AIDS

103. Which of the following statements is **INCORRECT** regarding malaria infection ?

(1) The infective stage for man is sporozoite

(2) Sporozoite firstly attacks the RBCs, resulting in their rupture

(3) Fever usually recurrent after every 3-4 days

(4) Haemozoin causes shivering and high fever

104. Mark correctly matched option :

	Source	Effect of compound
(A)		– Effect on CVS
(B)		– Sedative and painkiller
(C)		– Hallucinogen

(1) A and B

(2) B and C

(3) A, B and C

(4) Only C

105. MALT is associated with :

(1) Skin

(2) Lining of urinogenital tract

(3) Lining of coelom

(4) Lining of blood vessels

106. Principle of vaccination is based on which characteristic of :

- (1) Specificity
- (2) Diversity
- (3) Memory
- (4) Discrimination among self and non self

107. Interferons are __ (a) __ which are secreted from __ (b) __ :-

- (1) (a)-Glycerides, (b)-Virus infected cells
- (2) (a)-Amino acid, (b)-Normal cells
- (3) (a)-Proteins, (b)-Normal cells
- (4) (a)-Proteins, (b)-Virus infected cells

108. Read the following statement

- (A) Interferons are secreted by virus-infected cells and protect infected cells from further viral infection
- (B) B lymphocytes produce antibodies in response to pathogens into blood to fight with them
- (C) Macrophages are mucus secreting cells that trap microbes entering in the body
- (D) IgA is present in colostrum in early days of lactation to protect infant from diseases

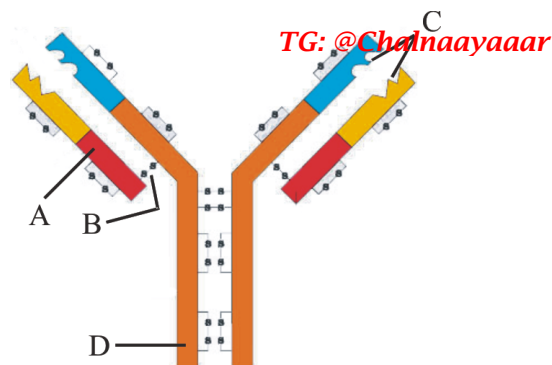
Choose the option having incorrect statement

- (1) A & D
- (2) A & C
- (3) B & D
- (4) Only C

109. HIV virus is:

- (1) Retrovirus which is unenveloped
- (2) Enveloped retrovirus which do not contain any protein capsid
- (3) Enveloped virus having only one molecule of reverse transcriptase
- (4) Enveloped retrovirus having two molecules of reverse transcriptase

110.



Identify (A), (B), (C), (D).

- (1) (A) Heavy chain (B) Disulphide bond (C) Epitope (D) Light chain
- (2) (A) Light chain (B) Disulphide bond (C) Antigen binding site (D) Heavy chain
- (3) (A) Light chain (B) Heavy chain (C) Antigen binding site (D) Disulphide bond
- (4) (A) Light chain (B) Disulphide bond (C) Heavy chain (D) Epitope

111. Which one of the following option gives the correct matching of a disease with its causative organism and mode of infection.

	Disease	Causative Organisms	Mode of Infection
(1)	Typhoid	<i>Salmonella typhi</i>	With inspirad air
(2)	Pneumonia	<i>Streptococcus pneumoniae</i>	Droplet infection
(3)	Elephantiasis	<i>Wuchereria bancrofti</i>	infected water and food
(4)	Malaria	<i>Plasmodium vivax</i>	Bite of male anopheles mosquito

112. A patient is suffering from high fever, stomach pain, Headache, Loss of appetite, Constipation intestinal ulcer. Which amongst following can be used as a diagnostic test :-

- (1) Widal Test (2) Elisa Test
(3) Montaux test (4) Shick test

113. How many of the following statement are false for transplantation and grafting :-

- (A) Coronary artery bypass grafting is the most common example of xenograft.
(B) Cornea transplant is successful and easy because cornea is avascular.
(C) Isograft is taken from genetically identical individual like identical twins.
(D) There is no requirement of MHC-matching in case of allograft.

- (1) Two (2) Three (3) One (4) Four

114. Which one of the following group is not of infectious disease :-

- (1) Cancer, Allergy, Rheumatoid arthritis
(2) Dysentery, Plague, Diphtheria
(3) Polio, Pneumonia, Tetanus
(4) Smallpox, Ringworm, Common cold

115. **Assertion:** Genital warts is a sexually transmitted disease

Reason: Genital warts spread through sexual intercourse with carriers of the disease

- (1) Both assertion & reason are true and reason is the correct explanation of assertion
(2) Both assertion & reason are true and reason is not the correct explanation of assertion
(3) Assertion is true but reason is false
(4) Both assertion and reason are false

116. Given below are two statements: TG: @Chalnaayaaar

Statement-I : Common examples of allergens are mites in dust, pollens, animal dander etc.

Statement-II : Substances to which exaggerated immune response is shown are called allergens

In light of the above statements, choose the most appropriate answer from the options given below :

- (1) Statement I and statement II are correct
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are incorrect
(4) Statement I is correct but statement II is incorrect

117. Read the following five statements (A–E) :

- (A) IgA antibody found in saliva, mucus, tear and colostrum
(B) Gonorrhoea, Syphilis, Genital herpes and Cancer are sexually transmitted disease
(C) Hepatitis-B vaccine is prepared by Recombinant DNA technology
(D) In Klinefelter syndrome total no. of Barr body is one
(E) In Artificial passive acquired immunity toxoids can be used.

How many of above statements are wrong

- (1) 4 (2) 3 (3) 2 (4) 1

118. Cancer is generally caused due to mutation of..... into.....and / or inactivation of.....

- (1) Oncogene, tumour suppressor gene, proto oncogene
(2) Tumor suppressor gene, oncogene, proto-oncogene
(3) Protooncogene, oncogene, tumour suppressor gene
(4) Apoptotic gene, tumor suppressor gene, protooncogene.

119. Along with lymphoid organs, WBCs are also present within the mucosa of digestive, respiratory and excretory tracts as MALT which constitutes about of total lymphoid tissue of the body:
(1) 50% (2) 75% (3) 15% (4) 30%
120. Which one is not included as at high risk of getting HIV infection -
(1) Individual who have multiple sexual partners.
(2) Drug addicts who take drugs orally.
(3) Individuals who require repeated blood transfusions.
(4) Children born to an HIV infected mothers.
121. Mark the odd one in an adult human w.r.t. lymphoid organs :-
(1) Peyer's patches (2) Spleen
(3) Liver (4) Thymus
122. Which one of the following vaccine is produced by recombinant DNA technology ?
(1) Polio (2) Small pox
(3) Hepatitis-B (4) Tuberculosis
123. Which of the following can be used to reduce the symptoms of Allergy ?
(a) Alanine (b) Histamine
(c) Steroid (d) Adrenaline
(1) a,b,c,d (2) b,c,d
(3) c,d (4) a,c,d
124. Which of the following disease is a disorder of blood of human ?
(1) Phenyl-Keto uria (2) Albinism
(3) Thalassemia (4) Color-blindness
125. Secretion of mammary gland contains which type of antibodies ? TG: @Chalnaayaaar
(1) I_gM (2) I_gD (3) I_gE (4) I_gA
126. Select the mismatched pair w.r.t. immunity.
(1) CMI→T-lymphocytes
(2) Passive immunity - ATS
(3) Hepatitis - B Vaccine - Killed vaccine
(4) Humoral-immunity-Immunoglobulin
127. Entamoeba histolytica is transmitted through :
(1) Droplet infection
(2) Vector borne
(3) Food and H_2O Contamination
(4) Insect bite
128. Disorder in which adenosine deaminase enzyme deficiency occur :-
(1) AIDS (2) SCID
(3) Rheumatoid arthritis (4) Phenylketouria
129. α and β thalassaemic cause due to mutation on chromosomal number respectively -
(1) 11,16 (2) 16,11
(3) 11,15 (4) 15,16
130. Select the options including are sexually transmitted disease.
(1) Typhoid, Cancer, AIDS
(2) AIDS, Malaria, Syphilis
(3) Gonorrhoea, AIDS, Syphilis
(4) AIDS, Genital herpes, Filaria

131. Select the *correct* match.
- (1) Sickle-Cell-anaemia - Autosomal Recessive
 - (2) Thalassemia – X - Linked Recessive
 - (3) Phenyl - ketouria – Autosomal dominant
 - (4) Albinism – X-linked
132. Which of the following is included in cellular barrier of innate-immunity ?
- (1) Monocyte (2) T-killer cell
 - (3) Plasma cells (4) Interterons
133. Active immunity is a immunity gained after -
- (1) Natural infection (2) Vaccination
 - (3) exposure to antigen (4) All of them
- 134.
- HIV-Virus

A

↓

Immuno-deficiency

B

↓

HIV factory

A & B respectively -
- (1) A → T_K Cell, B → T_H - Cell
 - (2) A → Macrophage, B → T_H - Cell
 - (3) A → T_K Cell, B → Macrophage
 - (4) A → T_H - Cell, B → Macrophage
135. Transformation of normal cells into cancerous cells may be induced by -
- (1) Physical agent (2) Chemical agent
 - (3) biological agent (4) all of them

SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

136. Cancer detection is based on. TG: @Chalnaayaaar
- (a) biopsy and histopathological studies of the tissues
 - (b) blood and bone marrow tests for increased cell counts in the case of leukemias
 - (c) radiography, computed tomography and magnetic resonance imaging
- (1) a and b (2) b and c
 - (3) a and c (4) a, b and c
137. Which immunoglobulin show specific linkage with basophill?
- (1) IgA (2) IgG (3) IgE (4) IgM
138. A non-infectious, unnatural and unusual reaction of a person to any substance or condition for which he is hypersensitive is termed as
- (1) Infection (2) Immunity
 - (3) Allergy (4) Toxin
139. Many diseases can be diagnosed by observing the symptoms in the patient. Which group of symptoms are indicative of pneumonia ?
- (1) Difficulty in respiration, fever, chills, cough, headache
 - (2) Constipation, abdominal pain, blood clots
 - (3) Nasal congestion and discharge, headache
 - (4) High fever, weakness, stomach pain, loss of appetite and constipation
140. Match the following :-

	Column-A		Column-B
A.	Fever	(i)	Cellular immunity
B.	Monocytes	(ii)	Humoral immunity
C.	Killer-T-cells	(iii)	Physiological barrier
D.	Plasma cells	(iv)	Cellular barrier

- (1) A-(iii) B-(i) C-(iv) D-(ii)
- (2) A-(iii) B-(iv) C-(i) D-(ii)
- (3) A-(ii) B-(iv) C-(i) D-(ii)
- (4) A-(ii) B-(i) C-(iv) D-(iii)

141. **Assertion** : The rupture of RBC is associated with toxin haemozoin.

Reason : Plasmodium attack RBC and cause fever.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

142. Given below are two statements :

Statement-I : Morphine extracted from latex of poppy plant.

Statement-II : Morphine is a very effective sedative and painkiller.

In light of the above statements, choose the most appropriate answer from the options given below :

- (1) Statement I and statement II are correct
- (2) Statement I is incorrect but statement II is correct
- (3) Both statement I and statement II are incorrect
- (4) Statement I is correct but statement II is incorrect

143. Consider the following statements -

I. Neural system and endocrine system influences our immune system.

II. Mind and mental state can affect our health.

III. Health is affected by genetic disorders, infections and life style.

IV. Health does simply mean only absence of disease.

Which of the above given statements are correct?

- (1) only I
- (2) I, II and IV
- (3) only III and IV
- (4) I, II and III

144. It is a chromosomal disorder in which loss of 'x' chromosome in female, leaving behind rudimentary ovaries with sterility.

- (1) Turner's syndrome
- (2) Down syndrome
- (3) Klinefelter syndrome
- (4) Jacob - syndrome

145. In AIDS, opportunistic infections are especially caused by :-

- (1) Toxoplasma
- (2) Mycobacterium
- (3) Viruses and Fungi
- (4) All of these

146. Cyclosporin-A, used as immuno-suppression agents is produced from :-

- (1) Bacteria
- (2) Virus
- (3) Fungi
- (4) Protozoa

147. AIDS does not spread due to

- (1) Kissing
- (2) Body fluids
- (3) Physical touch
- (4) Both 1 and 3

148. Which of the following STD not cause by bacteria ?

- (1) Syphilis
- (2) Hepatitis-B
- (3) Gonorrhoea
- (4) Chlamydiasis

149. AIDS is caused by HIV that infect -

- (1) T-helper cells
- (2) T-killer cells
- (3) B-lymphocyte
- (4) Plasma-cell

150. Housefly act as mechanical carrier in which of the following disease ?

- (1) Chikun-gunya
- (2) Filariasis
- (3) Amoebiasis
- (4) Pneumonia

Topic : HUMAN HEALTH & DISEASE, ECOLOGY (UPTO INTERSPECIFIC INTERACTIONS)

SECTION-A

Attempt All 35 questions

151. Which of the following is a pioneer community in hydrosere ?

- (1) Rooted submerged stage
- (2) Forest stage
- (3) Reed swamp stage
- (4) Phytoplankton stage

152. Mycorrhizae are the examples of :-

- (1) Amensalism (2) Parasitism
- (3) Mutualism (4) Commensalism

153. Cattle or goat are never browsing on *Calotropis* growing in abandoned field because of the presence of :-

- (1) Alkaloid like Quinine
- (2) Strychnine
- (3) Cardiac glycosides
- (4) Opium/Morphine

154. Primary succession and secondary succession take place on respectively :-

- (1) Newly created pond and Abandoned farm land
- (2) Newly created pond and bare rock
- (3) Deforested site and abandoned farm land.
- (4) Degraded forest and bare rock

155. The final community in a habitat is called:

- (1) Climax (2) Pioneer
- (3) Seral (4) Endemic

156. A forest community has been destroyed due to fire. After long time new community is again developed on this area it is :-

- (1) Primary succession
- (2) Secondary succession
- (3) Autogenic succession
- (4) Heterotrophic succession

157. Entire sequence of communities that successively change in a given area is :-

- (1) Seral communities
- (2) Climax communities
- (3) Sere
- (4) Pioneer communities

158. Which one area is not an example of primary succession?

- (1) Abandoned farmland
- (2) Newly formed lake
- (3) Soil of volcanic lava
- (4) Newly formed rock

159. An association of individual of same species living in the same habitat and interacting with each other is :-

- (1) Ecosystem (2) Biotic community
- (3) Population (4) Ecological niche

160. Which of the following means was used by warblers to avoid competition and coexistence ?

- (1) Difference in foraging activities
- (2) Habitat fragmentation
- (3) Competitive release
- (4) All of these

161. Community is defined as aggregation of :-

- (1) Individuals of the same kind
- (2) Individuals of same species
- (3) Individuals of a population
- (4) Populations of different species.

162. Habitat includes :-

- (1) Abiotic components
- (2) Biotic components
- (3) Biotic and abiotic components
- (4) Physiochemical components

163. **Assertion :** Most spectacular and evolutionary fascinating examples of mutualism are found in plant - Animal relationship.

Reason : Orchids show bewildering diversity of floral patterns.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

164. Connell's elegant field experiments showed that on rocky sea coasts of Scotland, the larger and competitively superior barnacle _____ dominates the intertidal area and exclude smaller barnacle _____ from that zone

- (1) Chathamalus, Balanus
- (2) Balanus, Balanus
- (3) Chathamalus, Chathamalus
- (4) Balanus, Chathamalus

165. Most successful parasites are those which do not

- (1) Grow free
- (2) Kill their host
- (3) Reproduce sexually
- (4) Survive in soil

166. **Assertion :** In an area, if there are 200 *Parthenium* plants and one huge banyan tree with large canopy, biomass is more meaningful measurement of N (Population Density).

Reason : In this case, number is meaning less which underestimates the enormous role of banyan tree in that community.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

167. Assertion : Under a particular set of selection pressure, organism evolve toward the most efficient reproductive strategy.

Reason : Some organism breed only once in their life while other breed many times during their lifetime.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

168. Resource partitioning is an important force which promotes

- (1) Coexistence
- (2) Antibiosis
- (3) Competitive release
- (4) Competition exclusion

169. Ecology is basically concerned with how many levels of biological organisation?

- (1) 4 (2) 3 (3) 2 (4) 1

170. Ecology at the A level is essentially B ecology.

Fill in the blanks with appropriate answer?

- (1) A-Community, B-Physiological
- (2) A-Organismic, B-Physiological
- (3) A-Community, B-Behavioural
- (4) A-Organismic, B-Population

171. How many of the following key elements characterize the habitat of an organism?

Temperature, water, predators, competitors, light, soil, parasites

- (1) 2 (2) 3 (3) 4 (4) 7

172. Niche of an organism is defined as -

- (1) Invariably defined range of conditions that an organism can tolerate.
- (2) Diversity in the resources it utilises.
- (3) Distinct functional role of organism in the ecological system.
- (4) All of these.

173. Which factor increases population size?

- (1) Food limitation
- (2) Adverse climate
- (3) Immigration
- (4) High predation pressure.

174. When resources in the habitat are unlimited, then the population grows in which fashion?

- (1) Exponential (2) Geometric
- (3) Logistic (4) Both 1 and 2

175. If there are 80 fruit flies in a cage, in about two weeks 8000 new fruit flies were born. Calculate birth rate?

- (1) 100 fruit flies per fruit fly per week.
- (2) 100 fruit flies per population per week.
- (3) 50 fruit flies per fruit fly per week.
- (4) 50 fruit flies per population per week.

176. Which of the following is a feature of population NOT of organism?

- (1) Birth (2) Age
(3) Growth rate (4) Death

177. In the given equation, $\frac{dN}{dt} = r N \left(\frac{K - N}{K} \right)$, what does K value denote?

- (1) Fitness
(2) Intrinsic rate of natural increase
(3) Carrying capacity
(4) Growth rate

178. How was the invasive Prickly pear Cactus was brought under control in Australia?

- (1) By introducing fungus that infects Cactus.
(2) Eradicating the pollinator of Cactus.
(3) Spraying weedicide.
(4) Introducing Cactus feeding moth.

179. Biological pest control methods are based on.

- (1) Predation (2) Commensalism
(3) Mutualism (4) Anensalism

180. What is the name of important predator is rocky intertidal communities of American pacific coast?

- (1) *Balanus* (2) *Pisaster*
(3) Shark (4) Seal

181. Which of the following interaction is evolved in most taxonomic groups from plants to higher vertebrates?

- (1) Parasitism (2) Commensalism
(3) Amensalism (4) Protocooperation

182. Which of following is parasite on Marine fish?

- (1) Copepods (2) Ticks
(3) Leech (4) Lice

183. The female wasp uses which part of fig tree as oviposition site?

- (1) Locule of ovary (2) Male flower
(3) Fruit (4) Sepals

184. What benefit does clown fish get as commensal from its host sea anemone?

- (1) Protection from parasites
(2) Protection from predators
(3) Food/Nutrients
(4) Immunity against pathogens

185. A population growing exponentially doubles within a year. Calculate intrinsic rate of natural increase?

- (1) 0.231 (2) 0.693 (3) 0.521 (4) 0.811

SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

186. The reason for prickly pear cactus to spread in Australia is :-

- (1) The availability of water
(2) Human activity
(3) Absence of cactus feeding animals
(4) Availability of predator in Australia

187. All of the following are examples of commensalism except :-

- (1) Orchid on mango branch
- (2) Cattle egret bird and grazing cattle
- (3) Sea anemone and clown fish
- (4) Cuckoo and Crow

188. Which of the following organism is highly distasteful to its predator because of a special chemical present in its body :-

- (1) *Pisaster* starfish
- (2) Prickly pear cactus
- (3) Monarch butterfly
- (4) *Parthenium*

189. Which of the following illustrates an example of Inter kingdom interaction ?

- (1) An actinomycetes Secreting antibiotic that kills nearby bacteria
- (2) An epiphyte growing on branch of tree
- (3) Lichen
- (4) Hermit crab carrying sea anemone to distant locations

190. In which of the following example one species help the other in some vital activity ?

- (1) Kangaroo rat - Fig tree
- (2) Lion - Deer
- (3) Pollinator - Insect
- (4) Flamingo Bird - Fishes

191. The current value of intrinsic growth (r) of a country is 1.1% then in how many years the population of the country will be doubled :-

- (1) 43 years
- (2) 58 years
- (3) 33 years
- (4) 63 years

192. A hunter tries to hunt a bird situated on a branch of tree. As he tries to stretch an arrow, a scorpion bites him. As a result, his arrow narrowly misses the bird and hits the monkey. Select the incorrect option :-

- (1) Scorpion - hunter - Amensalism
- (2) Scorpion - Bird - Commensalism
- (3) Hunter - Monkey - Amensalism
- (4) Monkey - Bird - Competition

193. Choose incorrect statement from following?

- (1) Group of individuals resulting from sexual and asexual reproduction are considered as population for the purpose of ecological studies.
- (2) It is at population level that natural selection operates to evolve desired traits.
- (3) If a new habitat is just being colonised, immigration may contribute more significantly to population growth than birth rates.
- (4) The r value of Norway rat was higher than that of r value of Indian population in 1981.

194. If $N = 1000$, $b = 0.7$, $d = 0.2$, find dN/dt ?

- (1) 700
- (2) 500
- (3) 200
- (4) 100

195. If pre-reproductive individuals are much less than reproductive individuals, then what is the shape of the age pyramid for this population.

- (1) Declining
- (2) Stable
- (3) Expanding
- (4) Growing

196. A survey was done in small town. Till last year total population = 5000

Natality = 20

Mortality = 5

Number of persons immigrated = 15

Number of persons emigrated = 10

Calculate the total population of the small town in the present year?

- (1) 5020 (2) 5015
- (3) 5030 (4) 5025

197. What is the 'r' value of flour beetle?

- (1) 0.015 (2) 0.0205
- (3) 0.011 (4) 0.12

198. If a population is growing in a habitat with limited resources a plot of population density (N) in relation to time (t) results in which type of curve?

- (1) J-shaped (2) Sigmoid
- (3) Bell-shaped (4) T-shaped

199. Which of the following is NOT a special adaptation of parasites that evolved in accordance with their life styles?

- (1) Loss of necessary sense organs.
- (2) Loss of digestive system.
- (3) Presence of adhesive organs or suckers to cling on to host.
- (4) High reproductive capacity.

200. In a forest, 20 rabbits were caught and tagged. They were released after tagging was done. In another attempt, 10 rabbits were caught out of which 2 were the ones which have been previously tagged. Calculate number of rabbits in the forest?

- (1) 200 (2) 300 (3) 500 (4) 100

Read carefully the following instructions :

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1. Each candidate must show on demand his/her Allen ID Card to the Invigilator.
2. No candidate, without special permission of the Invigilator, would leave his/her seat.
3. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty.
4. Use of Electronic/Manual Calculator is prohibited.
5. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
6. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
7. The candidates will write the Correct Name and Form No. in the Test Booklet/Answer Sheet.

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